Evaluation runs: chain-replication (chain-replication)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
3,2	6	18	conforms	0.1	2	4 h, 250 GB	6	21	brain01, 2026-11-02
3,3	90	1260	conforms	0.7	4	4 h, 250 GB	100	4507	brain01, 2026-11-02

stateful

stateful: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
3,2	6	18	conforms	0.0	4	4 h, 250 GB	brain01, 2026-11-03
3,3	90	1260	conforms	0.2	4	4 h, 250 GB	brain01, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 4 4 fifo.

offline

offline: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	6	18	conforms	0.0	8	1 h, 250 GB	0.0	0.0	0.0	merge	brain01, 2026-11-03
3,3	90	1260	conforms	0.1	8	1 h, 250 GB	0.0	0.1	0.0	merge	brain01, 2026-11-03
3,4	2520	211680	conforms	4.9	28	48 h, 80 GB	0.7	4.8	0.1	merge	pinky24, 2026-11-08
4,3	90	2700	conforms	1.0	15	48 h, 80 GB	0.1	0.9	0.1	merge	pinky10, 2026-11-07
4,4	2520	831600	conforms	9.5	10	48 h, 250 GB	0.5	9.4	0.1	merge	brain03, 2026-11-07

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	6	18	conforms	5.0	168	4 h, 250 GB	0.0	5.0	0.0	merge	brain01, 2026-11-03
3,3	90	1260	conforms	9.2	371	4 h, 250 GB	0.0	8.9	0.3	merge	brain01, 2026-11-03
3,4	2520	211680	conforms	196.5	6628	48 h, 80 GB	0.7	196.4	0.1	merge	pinky24, 2026-11-08
4,3	90	2700	conforms	13.1	2398	48 h, 80 GB	0.1	13.1	0.0	merge	pinky10, 2026-11-07
4,4	2520	831600	conforms	1072.5	20066	48 h, 250 GB	0.5	1072.4	0.1	merge	brain03, 2026-11-07

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	6	18	conforms	6.4	154	4 h, 250 GB		brain01, 2026-11-03
3,3	90	1260	conforms	12.5	2504	4 h, 250 GB		brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,4	211680	211680	conforms	196.5	6639	48 h, 80 GB	196.4	4.8	0.1	merge	pinky24, 2026-11-08
4,3	2700	2700	conforms	13.1	2404	48 h, 80 GB	13.1	0.9	0.0	merge	pinky10, 2026-11-07

context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	18	18	conforms	5.0	168	4 h, 250 GB	5.0	0.0	0.0	merge	brain01, 2026-11-03
3,3	1260	1260	conforms	9.0	371	4 h, 250 GB	8.9	0.1	0.1	merge	brain01, 2026-11-03
4,4	831600	831600	conforms	1072.5	20072	48 h, 250 GB	1072.4	9.4	0.1	merge	brain03, 2026-11-07

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,4	211680	211680	conforms	196.5	6639	48 h, 80 GB	4.8	196.4	0.1	merge	pinky24, 2026-11-08
4,3	2700	2700	conforms	13.1	2404	48 h, 80 GB	0.9	13.1	0.0	merge	pinky10, 2026-11-07

context-free twins offline, Java port against the Rust implementation: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3,2	18	18	conforms	5.0	168	4 h, 250 GB	0.0	5.0	0.0	merge	brain01, 2026-11-03
3,3	1260	1260	conforms	9.3	371	4 h, 250 GB	0.1	8.9	0.3	merge	brain01, 2026-11-03
4,4	831600	831600	conforms	1072.5	20072	48 h, 250 GB	9.4	1072.4	0.1	merge	brain03, 2026-11-07

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	18	18	conforms	6.7	170	4 h, 250 GB		brain01, 2026-11-03
3,3	1260	1260	conforms	13.2	2510	4 h, 250 GB		brain01, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
3,2	18	18	conforms	3.7	171	4 h, 250 GB		brain01, 2026-11-03
3,3	1260	1260	conforms	7.7	635	4 h, 250 GB		brain01, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 4 4 fifo.